MILITARY SPECIFICATION

COPPER-CHROMIUM-ALLOY CASTINGS

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

* 1.1 This specification covers copper-chromium-alloy castings, sand or centrifugally cast, which have been solution heat-treated, quenched, and artificially aged to a minimum Brinell hardness of 89.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM E 8	Tension Testing of Metallic Materials
ASTM E 10	Brinell Hardness of Metallic Materials
ASTM E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B 342	Electrical conductivity by Use of Eddy Currents

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. The raw materials used shall be of a quality suitable to produce castings which comply with all the requirements of this specification.

* 3.2 Chemical composition. The chemical composition shall conform to the following chemical requirements:

Copper alloy No. 815:

Copper, %	98.00 min.
Chromium, %	0.40 - 1.5
Total other elements, %	0.50 max.

3.3 Heat treatment. Castings shall be heat-treated and artificially aged in such manner as to produce material which will conform to the requirements of this specification.

* 3.4 Mechanical properties. After heat treatment as specified in 3.3, the mechanical properties of the castings shall conform to the following requirements:

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Tensile strength, psi	40,000 min.
Yield strength,[1] psi	30,000 min.
Elongation in 2 inches, %	15 min.
Brinell hardness	89 min.
Rockwell "B" hardness	60 min.

3.5 Electrical conductivity.

* 3.5.1 The electrical conductivity after solution heat treatment and full aging shall not be less than 75 percent.

3.5.2 One hundred percent conductivity shall be defined as the conductivity of copper having a resistance of 0.15328 ohm per meter, gram at 20° centigrade (68° Fahrenheit).

3.5.3 Conductivity shall be determined as given in 4.2.3 on samples taken from heat-treated castings from each melt.

3.6 Dimensions.

3.6.1 The supplier shall be responsible for furnishing castings that can be laid out and machined to the finished dimensions, within the permissible variations, shown on the applicable drawings and that will conform to such gages as may be specified. Castings of excessive size or weight shall not be furnished.

3.6.2 Cast-to-size portions of castings shall conform to such dimensions and permissible variations as may be shown on the applicable drawings and shall not be warped or distorted in such a manner as to adversely affect their appearance or serviceability.

3.7 Workmanship. Castings shall be of uniform quality and condition, free from injurious blow holes and porosity, hard spots, shrinkage defects, cracks, and other injurious imperfections and, before inspection, shall be smooth and well cleaned by sandblasting, tumbling, chipping, or other process approved by the inspector.

[1] Stress necessary to produce an elongation of 0.50 percent under load.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105.

* 4.1.2 Lot.

* 4.1.2.1 Type I (static) castings. A lot shall consist of all the castings from one melt defined as:

(a) The entire melt from a single furnace charge weighing not more than 10,000 pounds used to pour one or more castings

(b) The combination of two or more furnace charges or crucible melts weighing not more than 10,000 pounds into a single ladle used to pour one or more castings.

* 4.1.2.2 Type II (centrifugal) castings. A lot shall consist of all castings produced from one melt except that, when similar castings from several melts weighing less than 500 pounds each in the rough are involved, a lot shall consist of not more than 5,000 pounds of rough castings of the same alloy, kind, and size submitted for inspection at the same time. When two or more furnaces or crucible melts are used to charge a ladle for a pouring, the castings produced therefrom shall be considered as coming from a single melt.

4.2 Test methods.

* 4.2.1 Tension tests. All tension tests shall be conducted in accordance with ASTM E 8. Two tests shall be made for each lot, except when test coupons are permitted (see 4.1.2.1 and 4.1.2.2), in which case one test shall be made. Yield strength, when required, shall be by the method specified in the appropriate table herein for the alloy ordered.

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* 4.2.2 Hardness tests. When required, hardness tests shall be conducted in accordance with ASTM E 10 or E 18.

* 4.2.3 Conductivity tests. When required, the electrical conductivity determination shall be performed in accordance with ASTM B 342 on five samples randomly selected from each lot. Failure of any of the samples to meet the requirements given in 3.5 shall be cause for rejection of the lot represented.

* 4.2.4 Chemical analysis. The samples selected in accordance with 4.1.1 shall be analyzed in accordance with method 111 or 112 of FED-STD-151. A single analysis of a composite sample may be made. In case of dispute, referee analysis shall be by method 111. If another method of analysis is desired, it shall be as agreed upon by the contractor and the procuring acitivity.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified in the order or contract, packaging and packing shall be in accordance with approved commercial packing methods, adequate to protect each casting from injury and to insure safe delivery at destination. The method used shall be commensurate with shipping at the lowest practicable commercial rates.

5.2 Labeling and marking. Unless otherwise specified, each package shall be labeled in accordance with MIL-STD-129 and shall contain the following information:

(a) Purchase order or contract number
(b) Title, number, and date of this specification
(c) Part No. (as specified on drawings)
(d) Gross, tare, and net weights
(e) Name and symbol (if any) of manufacturer.

6. NOTES

6.1 Intended use. Copper-chromium alloy is intended for applications requiring high electrical conductivity and higher mechanical strength than is attainable with copper.

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6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification, the part drawing numbers, quantity and weight of castings required, as well as exceptions to this specification, applicable drawings, and other documents.

* 6.3 Supersession data. This specification includes the requirements of MIL-C-19310(NOrd) dated 25 January 1956 which was issued in lieu of NAVORD OS 3276. When drawings etc. refer to these superseded documents, this specification applies.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
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Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-19310A (OS), Copper-Chromium-Alloy Castings

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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